

Hack2Skill Personal Journal Application - Enterprise Security Edition

A production-grade, secure journal application built with industry best practices for encryption, authentication, and data protection.

🛡️ Security Features

Authentication & Authorization

- ✅ OAuth 2.0 + JWT token-based authentication
- ✅ bcrypt password hashing (cost factor 12, ~200ms per hash)
- ✅ Account lockout protection (5 failed attempts → 30 min lockout)
- ✅ Token rotation on refresh
- ✅ CSRF protection on state-changing operations
- ✅ Role-based access control (RBAC)

Encryption

- ✅ AES-256-GCM encryption at rest for all journal entries
- ✅ User-specific keys derived from master key using HKDF-SHA256
- ✅ Authenticated encryption (detects tampering)
- ✅ TLS 1.3+ for all network communication
- ✅ HTTPS-only cookies with Secure, HttpOnly, SameSite flags

Database Security

- ✅ Row-level security (RLS) enforced at database level
- ✅ User data isolation - queries always filtered by user ID
- ✅ Parameterized queries (SQLAlchemy ORM)
- ✅ No direct password access at database level
- ✅ Soft deletes for GDPR compliance
- ✅ Immutable audit log (INSERT ONLY)

Compliance

- ✅ GDPR-compliant (right to access, right to deletion)
- ✅ OWASP Top 10 protections
- ✅ Comprehensive audit trail
- ✅ Data minimization principles
- ✅ User consent tracking

API Security

- ✅ Rate limiting (100/hour default, configurable per endpoint)
- ✅ Security headers (HSTS, CSP, X-Frame-Options, etc.)
- ✅ Input validation and sanitization
- ✅ CORS with whitelist
- ✅ Request ID tracing
- ✅ Comprehensive error logging

📁 Project Structure

...

```
.
├── README.md           # This file
├── ARCHITECTURE.md      # System design & threat model
├── DEPLOYMENT.md        # Production deployment guide
├── QUICKSTART.md        # Quick start guide
├── API.md              # API reference documentation
├──
├── Core Application
├── app.py               # Flask application factory
├── config.py            # Configuration management
├── security.py          # Security utilities (encryption, auth, hashing)
├── models.py            # Database models (User, JournalEntry, AuditLog)
├──
├── Configuration
├── .env.example         # Environment variables template
├── requirements.txt     # Python dependencies
├──
├── Testing
├── tests.py             # Comprehensive test suite (50+ tests)
├──
├── Docker
├── Dockerfile           # Container image definition
├── docker-compose.yml   # Multi-container setup
```

Requirement

Help me build a user-authenticated Personal Journal web application that uses a free LLM API and database to store user details ,profile,journals

****User Flow:****

1. The user arrives at the landing page and is prompted to Sign In.
2. After successful authentication, the user is taken to their private dashboard.
3. The dashboard allows the user to write multi-turn "journal entries" or "reflections" and converse with Gemini.
4. Gemini provides helpful summaries, brainstorming ideas, or reflections on the user's input.
5. All interactions (prompts and Gemini responses) are saved to the Firestore, isolated strictly to this specific user so that different users can't read each other's entries.
6. The user can view a history of their past entries.

##7. Additional Reqs(Not as per Base or MVP requirement)

Add two things to each entry: image attachments and a place ie location integrtaiion.

Photos

- A "Photos" strip under the entry editor: drag-and-drop or click to upload, thumbnail grid, click to view full size, hover to remove.
- Images are stored in private cloud storage under your user folder, so only you can see them. Each attachment row records which entry it belongs to, the file path, and the original name.
- Limits: images only (JPEG/PNG/WEBP/GIF/HEIC-rejected), 10MB per file, multiple files at once.
- Deleting an entry removes its attachment rows; the stored files are removed at the same time from the app.
- The sidebar list shows a small photo-count badge on entries that have images.

Place

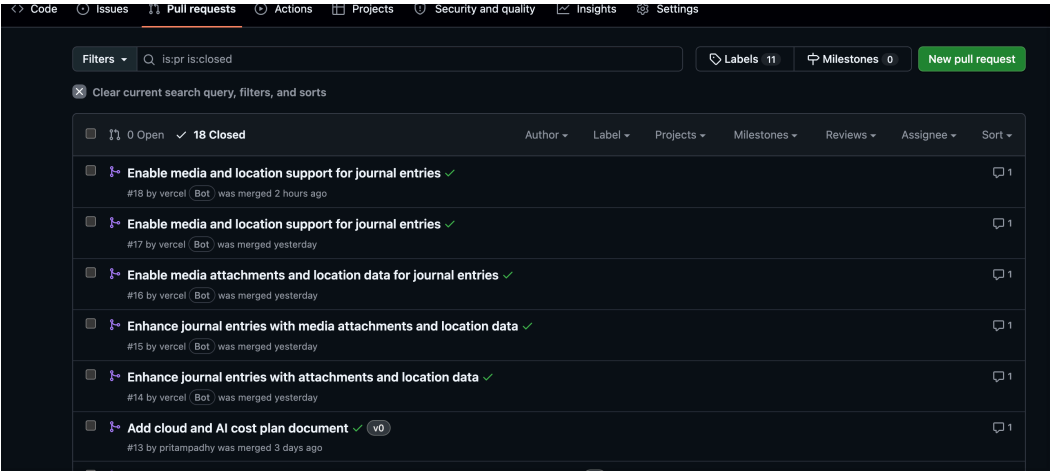
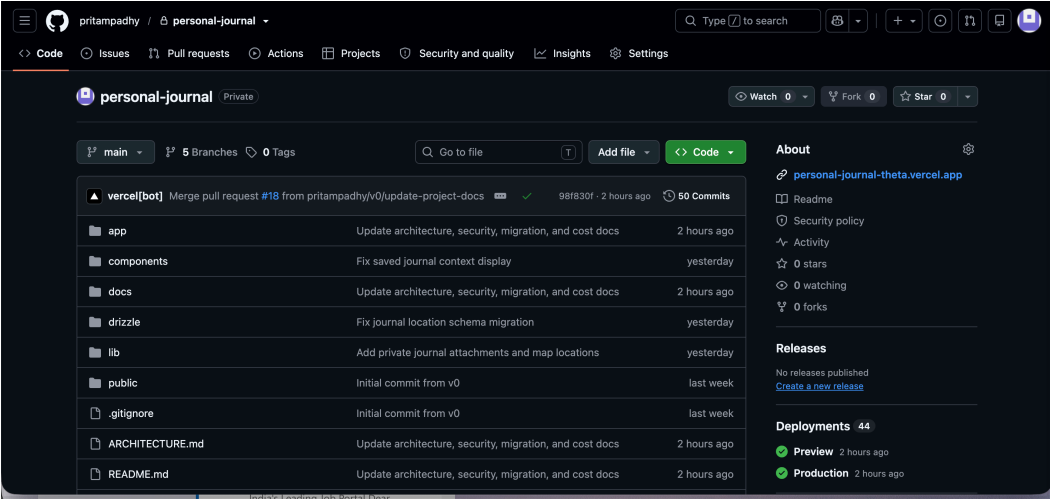
- A "Place" field in the entry header — you type a place name (e.g. "Lodhi Garden, Delhi"). Free text, no key needed.
- Once typed, the app looks the name up on OpenStreetMap's free geocoder and, if it finds it, saves coordinates and shows a small map preview (Leaflet + OpenStreetMap tiles) below the entry with a pin. If nothing is found, the text is still saved and no map shows.
- A place chip appears on entries in the sidebar list.
- The filter sidebar gains a place filter so you can narrow entries to a location; the existing keyword search also matches place names.

Technical notes

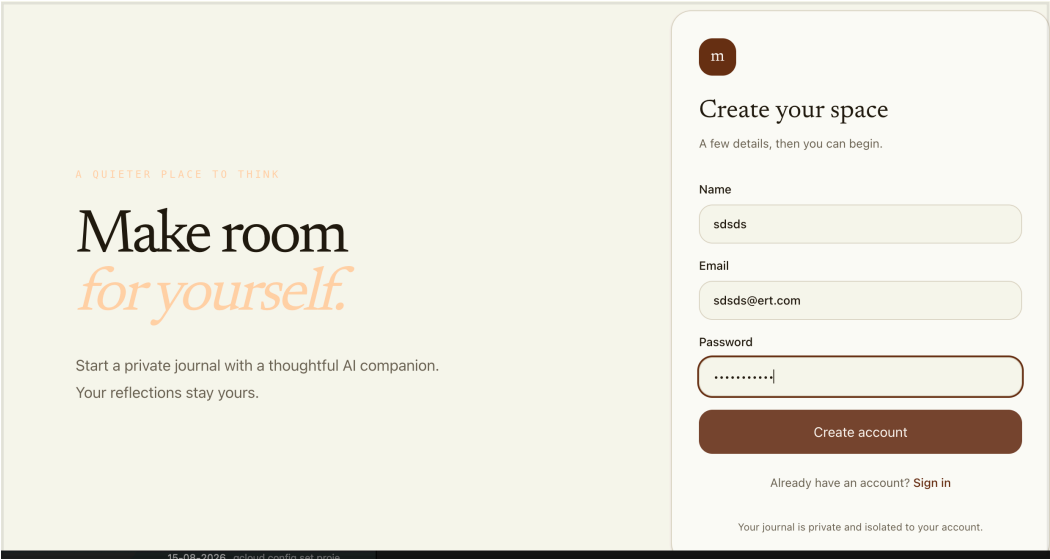
- Geocoding runs through a ``createServerFn`` (Nominatim, with a proper User-Agent and debounce/rate guard) so the browser never calls it directly and results can be cached per entry.
- Reflect context stays text-only (no images sent to the model), so the 300-token cap is unaffected.

Github. Code Base

Integrated PR creation and merging with branch main and feature
Zip file is attached codebase



Live Product Screenshots



Search entries...

September 2026

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MOOD

Great

Good

Okay

Low

Rough

TEMPLATE

Free write

Gratitude

Daily reflection

Dream journal

Travel

ENTRIES

TODAY'S REFLECTION

Free writeGratitudeDaily reflectionDream journalTravel

What happened today that you want to remember? How did it make you feel, and what did you learn?

I am writing a journal for a travel in michigan to google headquarters SF good journey in budget

Add context

Image or PDFChoose File No file chosen

Location labelPark, café, or place

Search entries...

September 2026

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Free write

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Dream journal

Travel

ENTRIES

Add context

Image or PDFChoose File No file chosen

Location labelDetroit Institute Of Arts

33.929710-117.412521Save context

Locations use OpenStreetMap. Enter coordinates from your map app.

Mood:

Save reflection

YOUR JOURNAL COMPANION

Summary: You shared a meaningful reflection about I am writing a journal for a travel in michigan to google headquarters SF good journey in budget

Insight: Naming this experience creates space to understand what matters most to you.

Question: What would feel like a helpful next step?

Your journal is private.

Search entries...

September 2026

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MOOD

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Good

Okay

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Rough

TEMPLATE

Free write

Gratitude

Daily reflection

Dream journal

Travel

ENTRIES

Image or PDFChoose File No file chosen

Park, café, or place

LatitudeLongitudeSave context

Locations use OpenStreetMap. Enter coordinates from your map app.

SAVED CONTEXT

Detroit Institute Of Arts (33.92971, -117.412521)

Mood:

Save reflection

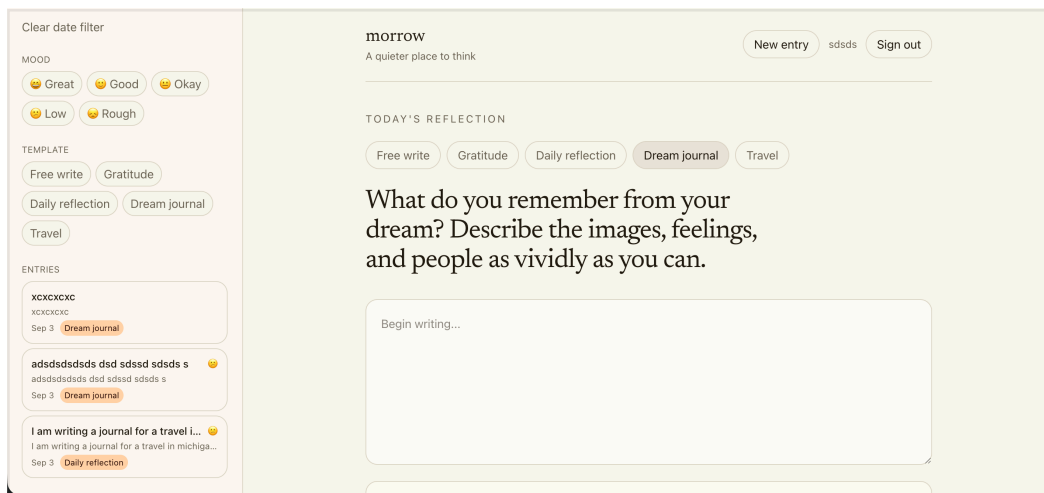
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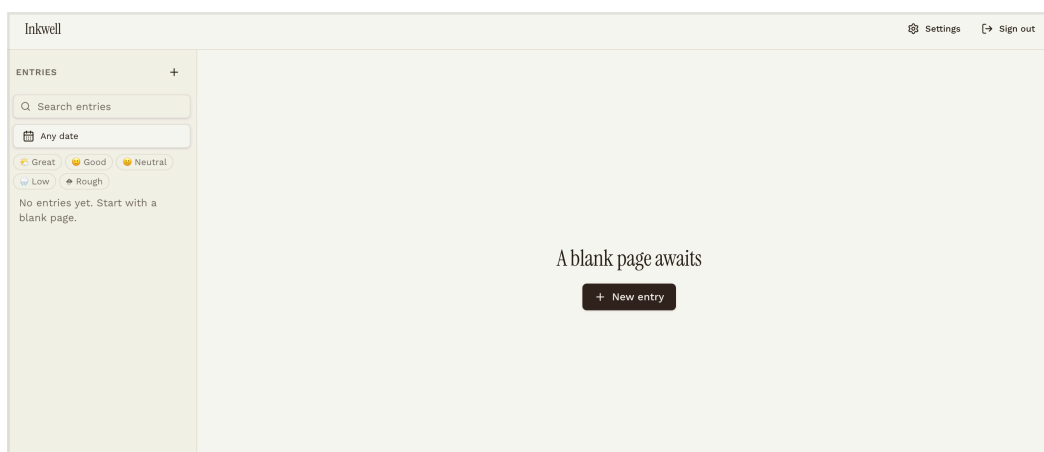
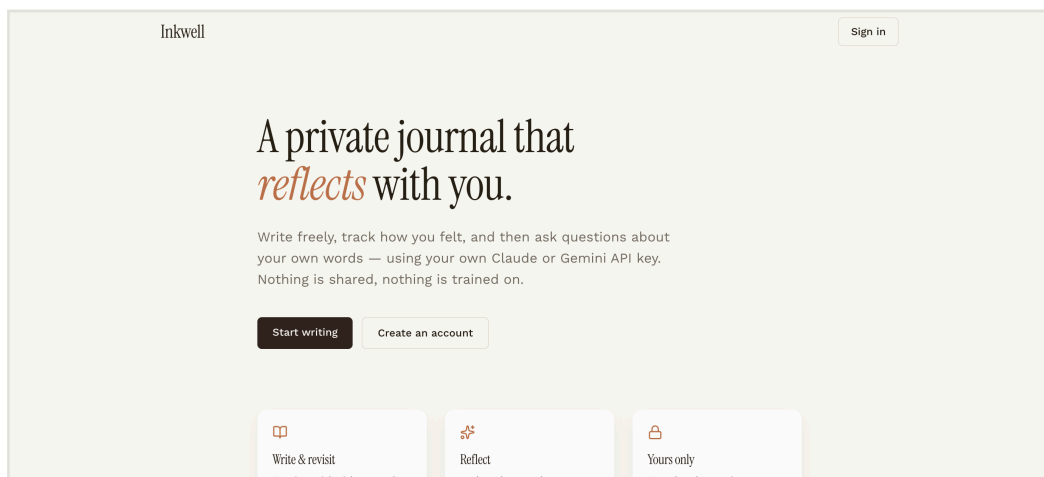
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2nd design



ENTRIES

+

Q Search entries

Any date

Great Good Neutral

Low Rough

Untitled

Sep 3, 2026

9/3/2026, 12:45:27 PM

Great

My musing for travel

📍

detroit institute of arts

A great day to travel from michigan to Google headquarters in SF on the way saw beautiful scenery and america midwest and other nice places

LOCATION

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Inkwell

Settings

Sign out

ENTRIES

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Q Search entries

Any date

Great Good Neutral

Low Rough

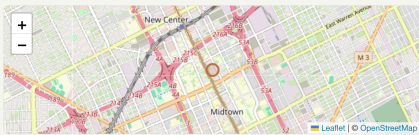
Untitled

Sep 3, 2026

LOCATION

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PHOTOS

Add photos

Drop images here, or click to choose (JPEG, PNG, WEBP, GIF - up to 10MB each)

25 words - unsaved

Save

Reflect

15-08-2026 notcloud config set proje

ENTRIES

+

Q Search entries

Any date

📍detroit institute of arts

Great Good Neutral

Low Rough

My musing for travel

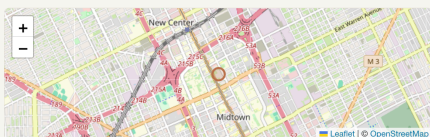
Sep 3, 2026

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LOCATION

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PHOTOS

Add photos



25 words - saved

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Reflect